DATE:13-10-2026 **CO1-AT2-SHORT ANSWERS**

NAME:VUTUKURI NAGA VISHNAV MAHESH **COURSE CODE: ITA0329**

REG NO : 192425141

1. 45" " MHz/(2×25" " kHz)= 900 channels

2. 560/7= 80 channels per cell

3. 100×8= 800 voice channels

4. 10log_10 (80/4)= 13.01 dB

5. -60-(-92)= 32 dB

6. (42" " MHz/300" " kHz)/7= 20 channels per cell

7. 64/2^4= 4 W

8. 2^2= 4 times

9. 4×15= 60 channels

10. f_d=vf/c= 83.33 Hz

11. 5/8= 0.625 ms

12. 24" " MHz/200" " kHz= 120 carriers

13. 10log_10 (8000)-33= 6.03 dBm

14. FSPL=32.44+20log_10 (1900)+20log_10 (2.5)= 105.97 dB

15. 43-(-87)= 130 dB

16. 150×2.5= 375 km²

17. (48" " MHz/200" " kHz)/6= 40 channels per cell

18. 150,000/6000= 25 cycles

19. 9/3= 3 handoffs (approximately)

20. -85-(-115)= 30 dB

21. 250×150" " kbps= 37.5 Mbps

22. 1500×4×(2.5/60)= 250 Erlangs

23. 250×12" " kHz= 3 MHz

24. 1500×600" " kbps= 900 Mbps

25. 350/(π×4^2)=6.96= 7 towers